Discussion on the K-Means Method

- Efficiency: O(tKn) where n: # of objects, K: # of clusters, and t: # of iterations
  - Normally, K, t << n; thus, an efficient method
- K-means clustering often terminates at a local optimal
  - Initialization can be important to find high-quality clusters
- Need to specify K, the number of clusters, in advance
  - There are ways to automatically determine the "best" K
  - In practice, one often runs a range of values and selected the "best" K value
- Sensitive to noisy data and outliers
  - Variations: Using K-medians, K-medoids, etc.
- K-means is applicable only to objects in a continuous n-dimensional space
  - Using the K-modes for categorical data
- Not suitable to discover clusters with non-convex shapes
  - Using density-based clustering, kernel K-means, etc.

Variations of K-Means

- Choosing better initial centroid estimates
  - K-means++, Intelligent K-Means, Genetic K-Means
- Choosing different representative prototypes for the clusters
  - K-Medoids, K-Medians, K-Modes
- Applying feature transformation techniques
  - Weighted K-Means, Kernel K-Means